

Jared Berry

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EDUCATION

Northwestern University | *McCormick School of Engineering*
Bachelor of Science in Mechanical Engineering | *Robotics Concentration*
Master of Science in Mechanical Engineering | *Robotics and Control*
Minor in Computer Science

Evanston, IL
September 2022 - June 2026
May 2025 - June 2026
September 2024 - June 2026

- GPA: **3.735/4.00**
- Selected Coursework: Data Structures and Algorithms, Differential Equations, Circuit Design, Mechanics, Quadrotor Design, Mechatronics, CAD, Dynamic Systems, Robotic Manipulation, Machine Learning, Control Systems, State Estimation/Filtering

EXPERIENCE

HAND ERC

Intelligent Dexterity Research Fellow

Evanston, IL
June 2025 – Present

- Awarded National Science Foundation research grant as part of the Research Experience for Undergraduates program.
- Enhanced the fidelity of a **physically realistic** task environment using the Drake robotics simulator with a ROS2 interface.
- Created pipeline for **reinforcement learning** in Drake, including a custom **Soft Actor Critic (SAC)** implementation.
- Validated RL pipeline with a benchmark **dexterous manipulation** task, and designed sparse and shaped reward functions.
- Prepared pipeline for use in **Sim-and-Real Co-Training**, and tuned model hyperparameters to improve **sim-to-real transfer**.

Shirley Ryan AbilityLab

Clinical ML Research Intern

Chicago, IL
June 2024 – July 2025

- Collected and **processed raw sensor data** from 9 DOF IMU sensors using Python.
- Extracted and selected features for use in an ML model using multiple **feature selection** algorithms.
- Created **Python ML model framework**, and predicted upper-limb motor function for recovering stroke patients ($R^2 = 0.93$).
- **Optimized model hyperparameters** using nested cross-validation and an Optuna hyperparameter search algorithm.
- **Nominated for best paper** at ICORR 2025 and presented findings to robotics professionals in a research symposium.

Mobile Mark, Inc

Engineering Intern

Itasca, IL
June 2022/2023 – September 2022/2023

- Fabricated active antennas for industrial mining and military applications, using **SolidWorks** and circuit CNC machines.

PROJECTS AND RESEARCH

Latent World Modeling from Images for Data Efficient Learning and Control

Interactive & Emergent Autonomy Lab

M.S. Thesis, Northwestern University
September 2025 - Present

- Conducted extensive literature review of state-of-the-art **latent world models**, and implemented key techniques.
- Developed and studied world model architectures, leveraging **convolutional VAEs** and **RNNs** to model complex dynamics.
- Exploring latent world models to enable **real-time control** of **embodied agents** in simulated and physical robotic systems.
- Producing a conference paper and Master's thesis with the scope and rigor of a journal publication.

Mobile Manipulation + Pick-and-Place

Capstone for ME449 Robotic Manipulation

Robotic Manipulation, Northwestern University
September 2024 - December 2024

- Generated a **dynamically calculated** reference trajectory with Python using the desired length and joint speed parameters.
- Created a **feedforward + PI feedback controller**, enabling task completion from an initial configuration with arbitrary error.
- Calculated **forward and inverse kinematics** for the mobile base and manipulator of a KUKA youBot.
- Prevented self-collisions and singularities with a custom function to improve motion quality and robot safety.

Rogers Research Group

Undergraduate Research Assistant

Northwestern University
September 2023 – June 2024

- Performed **micro-soldering** for wearable and implantable ECG, and also assisted in signal tuning and encapsulation.
- Collected and analyzed EEG/EOG data, and processed signals using Python in preparation for machine learning.
- Investigated the creation of a **ROS/Python** control framework for a robotic arm with EOG signal processing.
- Developed a **ROS2 position control framework** to manipulate a 7-DOF robotic arm.

PUBLICATIONS

- [1] S. Okita, **J. Berry**, R. Khazanchi, F. Lanotte, M. K. O'Brien and A. Jayaraman, "Machine Learning-Based Estimation of Upper Extremity Function in Stroke Rehabilitation Using Body-Worn Inertial Sensors," *International Conference On Rehabilitation Robotics (ICORR)*, 2025. [Accepted, Nominated for Best Paper]

TECHNICAL SKILLS AND INTERESTS

Languages: Python, C, C++, C#, x86-64 Assembly, Matlab, Java, Arduino, Racket

Software: Gymnasium, MuJoCo, Stable Baselines, ROS/ROS2, Linux, Drake (robotics simulator), Wandb, Unity, Godot, Inkscape

Machine Learning: World Models, RL, PyTorch, SAC, VAEs, RNNs, CNNs, SLAM, Optimization, Clinical ML

Fabrication and Design: SolidWorks, NX, Blender, CNC, Soldering, Micro-Soldering, Microcontrollers